

Hyperledger Fabric Node.js Chaincode Setup

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One-Time Setup Commands

1. Install system prerequisites

```
sudo apt update && sudo apt upgrade -y  
sudo apt install curl git docker.io docker-compose nodejs npm -y  
sudo systemctl enable docker  
sudo systemctl start docker  
sudo usermod -aG docker $USER  
newgrp docker
```

2. Download Fabric binaries & samples

```
curl -sSL https://bit.ly/2ysbOFE | bash -s
```

3. Add binaries to PATH (permanent)

```
echo 'export PATH=$PATH:$HOME/fabric-samples/bin' >> ~/.bashrc  
source ~/.bashrc
```

4. Create Node.js chaincode project folder

```
mkdir ~/my-chaincode  
cd ~/my-chaincode  
npm init -y
```

5. Install Node.js Fabric dependencies

```
npm install fabric-contract-api fabric-shim
```

Repeatable Commands (Per Session / Project Update)

1. Start Fabric test network

```
cd ~/fabric-samples/test-network
```

```
./network.sh up createChannel -c mychannel -ca
```

2. Stop Fabric network

```
./network.sh down
```

3. Deploy Node.js chaincode

```
./network.sh deployCC -ccn mycc -ccp ~/my-chaincode -ccl javascript
```

4. Initialize chaincode (if using initLedger)

```
peer chaincode invoke -o localhost:7050 --isInit \
```

```
--tls                                                                    --cafile
```

```
${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem \
```

```
--channelID mychannel \
```

```
--name mycc \
```

```
--peerAddresses localhost:7051 \
```

```
--tlsRootCertFiles
```

```
${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/certificate \
```

```
--waitForEvent \
```

```
-c '{"Args":["initLedger"]}'
```

5. Invoke chaincode transaction

```
peer chaincode invoke -C mychannel -n mycc -c '{"Args":["createAsset","asset1","Ali","200"]}'
```

6. Query chaincode transaction

```
peer chaincode query -C mychannel -n mycc -c '{"Args":["readAsset","asset1"]}'
```

Summary Table

Step	Command	Notes
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Install system tools | `sudo apt install ...` | One-time

Download Fabric | `curl -sSL ...` | One-time

Add PATH | `echo 'export PATH=...'` | One-time

Create chaincode | `mkdir ~/my-chaincode` | One-time per project

Install Node deps | `npm install fabric-contract-api fabric-shim` | One-time per project

Start network | `./network.sh up ...` | Repeat as needed

Stop network | `./network.sh down` | Repeat as needed

Deploy chaincode | `./network.sh deployCC ...` | Repeat on chaincode updates

Invoke/query | `peer chaincode invoke/query` | Repeat as needed